

# A pencil-and-paper encryption system based on the Enigma G (Abwehrenigma)

Step 1:

BSCISSAA  
IDBYROWS

Settings: Rotor B  
I → O

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD

|   |   |         |           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---------|-----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | A | KRYPTOS | ABCDEFGHI | G | H | I | J | K | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | R |
| O | B | RYPTOS  | ABCDEFGHI | H | I | J | K | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | E |
| T | C | YPTOS   | ABCDEFGHI | I | J | K | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | F |
| O | D | PTOS    | ABCDEFGHI | J | K | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | L |
| R | E | TOS     | ABCDEFGHI | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | E |
|   | F | OS      | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | C |
|   | G | S       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | T |
|   | H | A       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | O |
|   | I | B       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | R |
|   | J | C       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | K | D       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | L | E       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | M | F       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | N | G       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | L |
|   | O | H       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | P | I       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | Q | J       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | R | L       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | S | M       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | T | N       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | U | Q       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | V | U       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | W | V       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | X | W       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | Y | X       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |
|   | Z | Z       | ABCDEFGHI | J | L | M | N | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | ? |

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD

Step 2:

SCISSA  
UUNDT0

Settings: Reflector S  
O → U

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD

|   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | A | KRYPTOS | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |   |
| O | B | RYPTOS  | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
| T | C | YPTOS   | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
| O | D | PTOS    | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
| R | E | TOS     | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
|   | F | OS      | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
|   | G | S       | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |
|   | H | A       | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |
|   | I | B       | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |
|   | J | C       | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |
|   | K | D       | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |
|   | L | E       | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |
|   | M | F       | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |
|   | N | G       | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |
|   | O | H       | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |
|   | P | I       | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |   |
|   | Q | J       | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |   |   |
|   | R | L       | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |   |   |   |   |
|   | S | M       | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |   |   |   |   |   |
|   | T | N       | O | P | Q | R | S | T | U | V | W | X | Y | Z | ? |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|   | U | Q       | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |   |
|   | V | U       | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |   |
|   | W | V       | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |   |
|   | X | W       | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |   |
|   | Y | X       | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |   |
|   | Z | Z       | Y | X | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |   |

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD

Step 3:

BSCISSA  
USEDTHE

Settings: Rotor B  
U → U

ABCDEFGHIJKLMNOPQRSTUVWXYZ N OPQRSTUVWXYZABCD

|   |   |                  |   |                   |   |   |
|---|---|------------------|---|-------------------|---|---|
| R | A | KRYPTOSABCDEFGHI | G | HIJLMNQUVWXXZKRYP | S | R |
| O | B | RYPTOSABCDEFGHI  | H | IJLMNQUVWXXZKRYPT | ? | E |
| T | C | YPTOSABCDEFGHI   | I | JLMNQUVWXXZKRYPTO | ? | F |
| O | D | PTOSABCDEFGHI    | J | LMNQUVWXXZKRYPTOS | ? | L |
| R | E | TOSABCDEFGHIJ    | L | MNQUVWXXZKRYPTOSA | ? | E |
|   | F | OSABCDEFGHIJL    | M | NQUVWXXZKRYPTOSAB | ? | C |
|   | G | SABCDEFGHIJLM    | N | QUVWXXZKRYPTOSABC | ? | T |
|   | H | ABCDEFGHIJLMN    | Q | UVWXXZKRYPTOSABCD | ? | O |
|   | I | BCDEFGHIJLMNQ    | U | VWXXZKRYPTOSABCDE | ? | R |
|   | J | CDEFGHIJLMNQU    | V | WXXZKRYPTOSABCDEF | ? |   |
|   | K | DEFGHIJLMNQUV    | W | XZKRYPTOSABCDEFGH | ? |   |
|   | L | EFGHIJLMNQUVW    | X | ZKRYPTOSABCDEFGH  | ? |   |
|   | M | FGHIJLMNQUVWX    | Z | KRYPTOSABCDEFGHI  | ? |   |
|   | N | GHIJLMNQUVWXZ    | K | RYPTOSABCDEFGHIJ  | L |   |
|   | O | HIJLMNQUVWXZK    | R | YPTOSABCDEFGHIJL  | ? |   |
|   | P | IJLMNQUVWXZKR    | Y | PTOSABCDEFGHIJLM  | ? |   |
|   | Q | JLMNQUVWXZKRY    | P | TOSABCDEFGHIJLMN  | ? |   |
|   | R | LMNQUVWXZKRYP    | T | OSABCDEFGHIJLMNQ  | ? |   |
|   | S | MNQUVWXZKRYPT    | O | SABCDEFGHIJLMNQU  | ? |   |
|   | T | NQUVWXZKRYPTO    | S | ABCDEFGHIJLMNQUV  | ? |   |
|   | U | QUVWXZKRYPTOS    | A | BCDEFGHIJLMNQUV   | ? |   |
|   | V | UVWXXZKRYPTOSA   | B | CDEFGHIJLMNQUVW   | ? |   |
|   | W | VWXXZKRYPTOSAB   | C | DEFGHIJLMNQUVWX   | ? |   |
|   | X | WXZKRYPTOSABC    | D | EFGHIJLMNQUVWXZ   | ? |   |
|   | Y | XZKRYPTOSABCD    | E | FGHIJLMNQUVWXZK   | ? |   |
|   | Z | ZKRYPTOSABCDE    | F | GHIJLMNQUVWXZKRY  | ? |   |

ABCDEFGHIJKLMNOPQRSTUVWXYZ N OPQRSTUVWXYZABCD

Step 4:

PALIMPSESTPALIMPSESTPALIMPSESTPALIMPSESTPALIMPSESTPALIMPSELTPAL  
BETWEENSUBTLESHADINGANDTHEABSENCEOFFLIGHTLIESTHENUANCEOFIQLUSION

Settings: Rotor B → ?, Reflector S → L

If there is a notch (turnover) at rotor setting B, the reflector also turns depending on the plain text letter. This is, what the word ASTATOS is maybe refering to. Not sure how the rotor is turning. The code word SCHEIDT for the unknown alphabet on the right is just a guess.

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD

|    |   |         |              |   |         |       |                  |    |   |
|----|---|---------|--------------|---|---------|-------|------------------|----|---|
| R  | A | KRYPTOS | ABCDEF       | G | HIJLMNQ | UVWXZ | KRYPT            | S  | R |
| O  | B | RYPTOS  | ABCDEFG      | H | IJLMNQ  | UVWXZ | KRYPT            | C? | E |
| T? | C | YPTOS   | ABCDEFGH     | I | JLMNQ   | UVWXZ | KRYPTO           | H? | F |
| O  | D | PTOS    | ABCDEFGH     | I | JLMNQ   | UVWXZ | KRYPTOS          | E? | L |
| R  | E | TOS     | ABCDEFGHIJ   | L | MNQ     | UVWXZ | KRYPTOSA         | I? | E |
| F  |   | OS      | ABCDEFGHIJL  | M | NQ      | UVWXZ | KRYPTOSAB        | D? | C |
| G  |   | S       | ABCDEFGHIJLM | N | Q       | UVWXZ | KRYPTOSABC       | T? | T |
| H  |   | ABC     | DEFGHIJLMN   | Q | U       | VWXZ  | KRYPTOSABCD      | ?  | O |
| I  |   | BC      | DEFGHIJLMNQ  | U | V       | WXZ   | KRYPTOSABCDE     | ?  | R |
| J  |   | C       | DEFGHIJLMNQU | V | W       | XZ    | KRYPTOSABCDEF    | ?  |   |
| K  |   | DE      | FGHIJLMNQUV  | W | X       | Z     | KRYPTOSABCDEFG   | ?  |   |
| L  |   | E       | FGHIJLMNQUVW | X | Z       |       | KRYPTOSABCDEFGH  | ?  |   |
| M  |   | F       | GHIJLMNQUVWX | Z |         |       | KRYPTOSABCDEFGHI | ?  |   |
| N  |   | G       | HIJLMNQUVWXZ | K |         |       | RYPTOSABCDEFGHIJ | L  |   |
| O  |   | H       | IJLMNQUVWXZK | R |         |       | YPTOSABCDEFGHIJL | ?  |   |
| P  |   | I       | JLMNQUVWXZKR | Y |         |       | PTOSABCDEFGHIJLM | ?  |   |
| Q  |   | J       | LMNQUVWXZKRY | P |         |       | TOSABCDEFGHIJLMN | ?  |   |
| R  |   | L       | MNQUVWXZKRYP | T |         |       | OSABCDEFGHIJLMNQ | ?  |   |
| S  |   | M       | NQUVWXZKRYPT | O |         |       | SABCDEFGHIJLMNQU | ?  |   |
| T  |   | N       | QUVWXZKRYPTO | S |         |       | ABCDEFGHIJLMNQUV | ?  |   |
| U  |   | Q       | UVWXZKRYPTOS | A |         |       | BCDEFGHIJLMNQUVW | ?  |   |
| V  |   | U       | VWXZKRYPTOSA | B |         |       | CDEFGHIJLMNQUVWX | ?  |   |
| W  |   | V       | WXZKRYPTOSAB | C |         |       | DEFGHIJLMNQUVWXZ | ?  |   |
| X  |   | W       | XZKRYPTOSABC | D |         |       | EFGHIJLMNQUVWXZK | ?  |   |
| Y  |   | X       | ZKRYPTOSABCD | E |         |       | FGHIJLMNQUVWXZKR | ?  |   |
| Z  |   | Z       | KRYPTOSABCDE | F |         |       | GHIJLMNQUVWXZKRY | ?  |   |

ABCDEFGHIJKLM N OPQRSTUVWXYZABCD